



BCSE309L- Cryptography and Network Security

VELLORE INSTITUTE OF TECHNOLOGY, VELLORE

Slots: E1+TE1, E2+TE2

Subject Code: BCSE309L

Class ID: VL2024250501924, VL2024250501924 **Faculty:** Dr. S. M. Farooq

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DIGITAL ASSIGNMENT QUESTIONS

1. Does one time pad achieve perfect secrecy? Why or why not. **[2M]**
2. Use a Hill cipher to encrypt the message “We live in an insecure world”
Use the following key. **[3M]**

$$K = \begin{pmatrix} 03 & 02 \\ 05 & 07 \end{pmatrix}$$

3. The function f used in each round of DES is invertible. Yes or No?
Explain. **[2M]**
4. AES has a larger block size than DES (128 versus 64). Is this an advantage or disadvantage? Explain. **[1 M]**
5. In AES, the size of the block is the same as the size of the round key (128 bits); in DES, the size of the block is 64 bits, but the size of the round key is only 48 bits. What are the advantages and disadvantages of AES over DES with respect to this difference? **[2 M]**